

Algebra II with Trigonometry

Chapter 7.3

Mental Math (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:
[4592201_alg-2trig-h-chp-7-3-note-docx.pdf](#)

1. **Double Angle from Sine and Cosine** If $\sin x = \frac{3}{5}$ and $\cos x = \frac{4}{5}$, what is $\sin 2x$?
2. **Double Angle from Tangent** If $\tan x = \frac{1}{2}$, what is $\tan 2x$?
3. **Cosine Double-Angle Shortcut** If $\cos x = \frac{\sqrt{3}}{2}$, what is $\cos 2x$?
4. **Half-Angle Exact Value** What is $\cos \frac{\pi}{8}$?
5. **Half-Angle Tangent** What is $\tan 22.5^\circ$?
6. **Lowering a Power** Rewrite $\sin^2 x$ using only cosine.
7. **Another Power-Lowering Idea** Rewrite $\cos^2 x$ using only cosine.
8. **Product to Sum** Rewrite $\sin 3x \cos x$ as a sum.
9. **Sum to Product** Rewrite $\sin 5x + \sin x$ as a product.
10. **Quick Exact Evaluation** Find the exact value of $\cos 75^\circ + \cos 15^\circ$.